

Rational Functions: Asymptotes, Holes & Graphing

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Answer Key – Problems 1–10

Problem 1

$$f(x) = \frac{x-2}{x^2-4}$$

Hole

VA

HA: $y=0$

Factor: $\frac{(x-2)}{(x-2)(x+2)}$

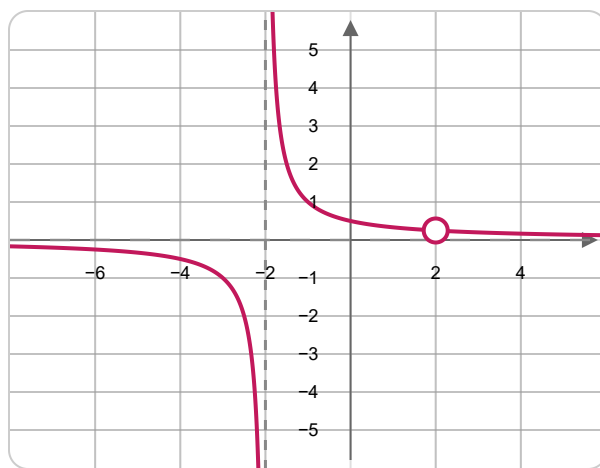
Hole: $(x-2)$ cancels $\rightarrow$ Hole at $x = 2$. Reduced: $\frac{1}{x+2} \cdot f(2) = \frac{1}{2+2} = \frac{1}{4} \rightarrow$ **Hole at**
 $\left(2, \frac{1}{4}\right)$

VA: $x+2 = 0 \rightarrow x = -2$

HA: $\deg(\text{num})=1 < \deg(\text{den})=2 \rightarrow y = 0$

Direction near $x = -2$: $x \rightarrow -2^+$: $(x+2) \rightarrow 0^+$, numerator = $1 > 0 \rightarrow +\infty$; $x \rightarrow -2^-$: $\rightarrow -\infty$

Hole: $\left(2, \frac{1}{4}\right)$ | **VA:** $x = -2$ | **HA:** $y = 0$



$f(x) = \frac{x-2}{x^2-4} - \text{reduced: } \frac{1}{x+2}, \text{ Hole } \left(2, \frac{1}{4}\right), \text{ VA } x = -2$



Problem 2

$$f(x) = \frac{x^2 - 9}{x + 3}$$

Hole

No VA

Linear after cancel

Factor: $\frac{(x - 3)(x + 3)}{(x + 3)}$

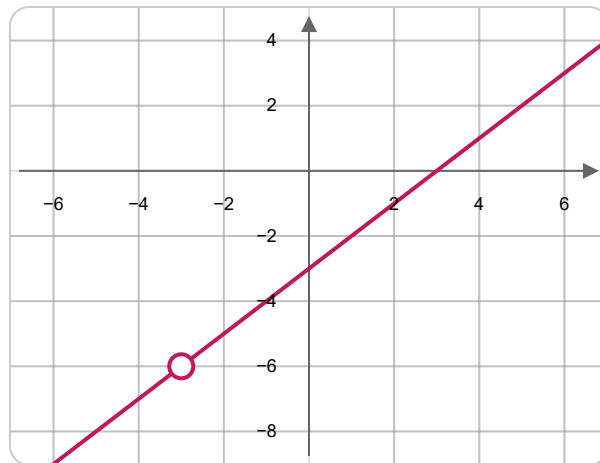
Hole: $(x + 3)$ cancels $\rightarrow$ Hole at $x = -3$. Reduced: $x - 3$. $f(-3) = -3 - 3 = -6 \rightarrow$ **Hole at** $(-3, -6)$

VA: No remaining denominator factor $\rightarrow$ **No VA**

HA / SA: Reduced form is the line $y = x - 3 \rightarrow$ **No HA. This is a line with a hole.** (oblique asymptote $y = x - 3$ if you must name one)

x-int: $x - 3 = 0 \rightarrow (3, 0)$

Hole: $(-3, -6)$ | **No VA** | **Simplified:** $y = x - 3$



Line $y = x - 3$ with hole at $(-3, -6)$

Problem 3

$$f(x) = \frac{2x + 6}{x^2 + 5x + 6}$$

Hole

VA

HA: $y=0$

Factor: $\frac{2(x+3)}{(x+3)(x+2)}$

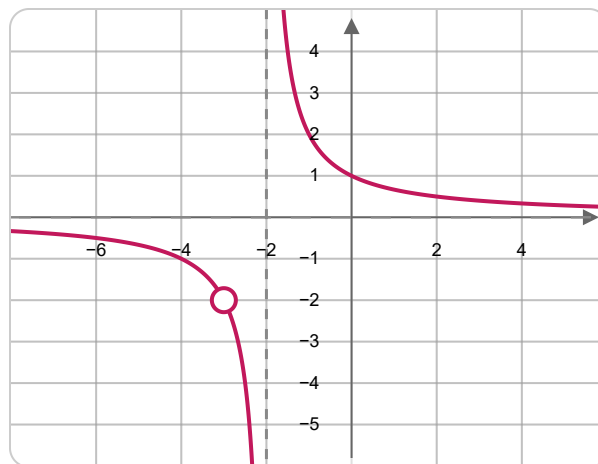
Hole: $(x+3)$ cancels $\rightarrow$ Hole at $x = -3$. Reduced: $\frac{2}{x+2} \cdot f(-3) = \frac{2}{-3+2} = \frac{2}{-1} = -2$
 $\rightarrow$ **Hole at** $(-3, -2)$

VA: $x+2=0 \rightarrow x = -2$

HA: $\deg(\text{num})=1 < \deg(\text{den})=2 \rightarrow y = 0$

y-int: $f(0) = \frac{6}{6} = 1 \rightarrow (0, 1)$

Hole: $(-3, -2)$ | **VA:** $x = -2$ | **HA:** $y = 0$



Reduced: $\frac{2}{x+2}$ — Hole $(-3, -2)$, VA $x = -2$

Problem 4 ★ Kenny's emphasis problem

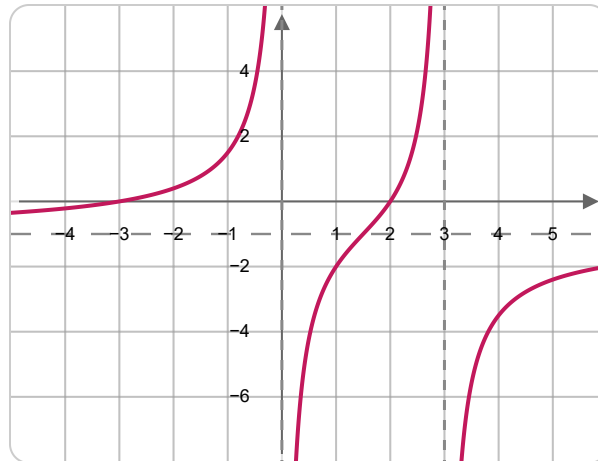
$$f(x) = \frac{x^2 + x - 6}{-x^2 + 3x}$$

Two VAs

No Hole

HA: $y = -1$ **Factor:** Numerator: $(x + 3)(x - 2)$ Denominator: $-x^2 + 3x = -x(x - 3)$ (factor out $-x$)

So: $\frac{(x + 3)(x - 2)}{-x(x - 3)}$

Hole check: No common factors between $(x + 3)(x - 2)$ and $-x(x - 3) \rightarrow$ **No holes****VA:** Denom = 0 when $x = 0$ or $x = 3 \rightarrow$ **VA:** $x = 0$ **and** $x = 3$ **HA:** $\deg(\text{num})=2 = \deg(\text{den})=2 \rightarrow$ ratio of leading coefficients: $\frac{1}{-1} = -1 \rightarrow$ **HA:** $y = -1$ **x-int:** Numerator = 0: $x = -3$ or $x = 2 \rightarrow (-3, 0)$ and $(2, 0)$ **Note:** The negative leading coefficient in the denominator flips the HA sign. Always factor out the negative carefully.**No hole | VA: $x = 0, x = 3$ | HA: $y = -1$** 

$$\frac{(x+3)(x-2)}{-x(x-3)} - \text{VAs } x = 0, x = 3, \text{ HA } y = -1$$

Problem 5

$$f(x) = \frac{3x^2}{x^2 - 1}$$

Two VAs

No Hole

HA: $y=3$

Factor: $\frac{3x^2}{(x-1)(x+1)}$ – numerator $3x^2$ has no factor matching denominator → **No holes**

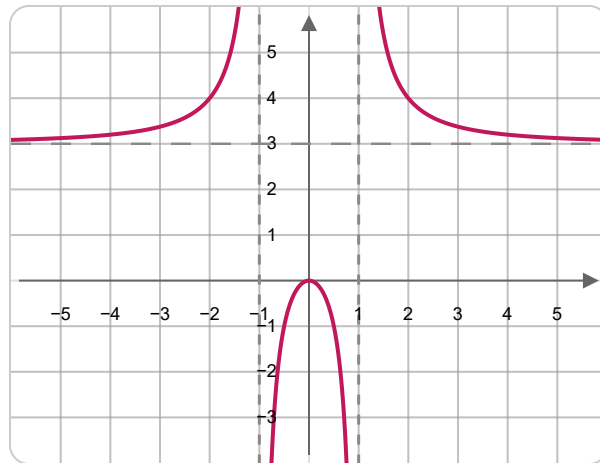
VA: $x = 1$ and $x = -1 \rightarrow$ **VA:** $x = -1$ **and** $x = 1$

HA: deg equal, leading coefficients $\frac{3}{1} \rightarrow$ **HA:** $y = 3$

x-int: $3x^2 = 0 \rightarrow x = 0$ (touches, doesn't cross) → $(0, 0)$

y-int: $f(0) = 0 \rightarrow (0, 0)$

No hole | VA: $x = -1, x = 1$ | HA: $y = 3$



$\frac{3x^2}{(x-1)(x+1)}$ – VAs $x = \pm 1$, HA $y = 3$, touches $(0, 0)$

Problem 6

$$f(x) = \frac{x^2 - 4}{x^2 - x - 6}$$

Hole

VA

HA: $y=1$

Factor: $\frac{(x-2)(x+2)}{(x-3)(x+2)}$

Hole: $(x+2)$ cancels $\rightarrow$ Hole at $x = -2$. Reduced: $\frac{x-2}{x-3}$. $f(-2) = \frac{-2-2}{-2-3} = \frac{-4}{-5} = \frac{4}{5} \rightarrow$

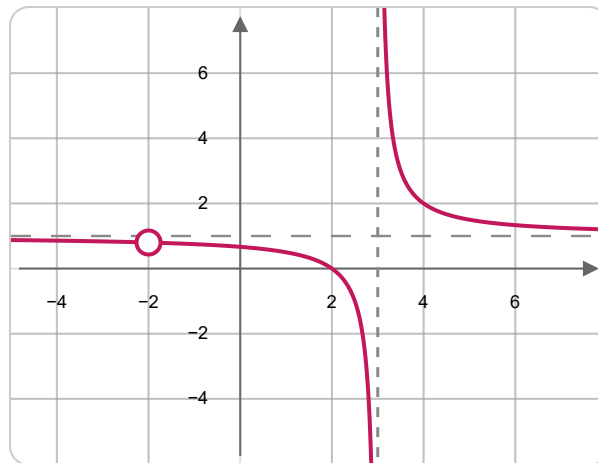
Hole at $\left(-2, \frac{4}{5}\right)$

VA: $x-3=0 \rightarrow x=3$

HA: deg equal, leading coefficients $\frac{1}{1} = 1 \rightarrow$ **HA:** $y=1$

x-int (reduced): $x-2=0 \rightarrow (2, 0)$

Hole: $\left(-2, \frac{4}{5}\right)$ | **VA:** $x=3$ | **HA:** $y=1$



Reduced: $\frac{x-2}{x-3}$ — Hole $\left(-2, \frac{4}{5}\right)$, VA $x=3$, HA $y=1$

Problem 7

$$f(x) = \frac{x^2 + 2x - 8}{x + 4}$$

Hole

No VA

Linear: $y=x-2$

Factor: $\frac{(x+4)(x-2)}{(x+4)}$

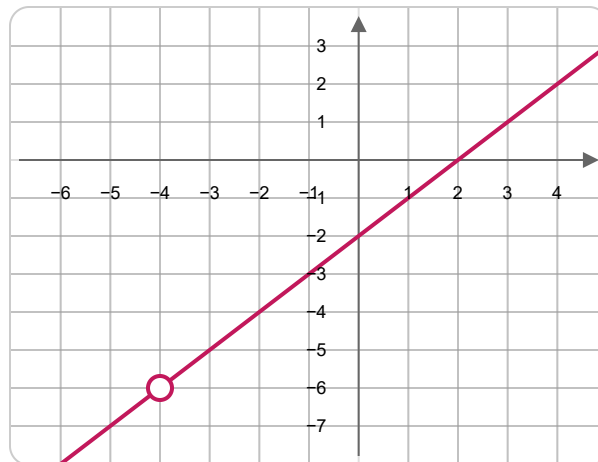
Hole: $(x+4)$ cancels $\rightarrow$ Hole at $x = -4$. Reduced: $x - 2$. $f(-4) = -4 - 2 = -6 \rightarrow$ **Hole at** $(-4, -6)$

VA: No remaining denominator $\rightarrow$ **No VA**

HA / SA: Simplified to a line $y = x - 2 \rightarrow$ **No HA. Line with hole.**

x-int: $x - 2 = 0 \rightarrow (2, 0)$

Hole: $(-4, -6)$ | **No VA** | **Simplified:** $y = x - 2$



Line $y = x - 2$ with hole at $(-4, -6)$

Problem 8

$$f(x) = \frac{x^2 - 1}{x - 1}$$

Hole

No VA

Linear: $y=x+1$

Factor: $\frac{(x-1)(x+1)}{(x-1)}$

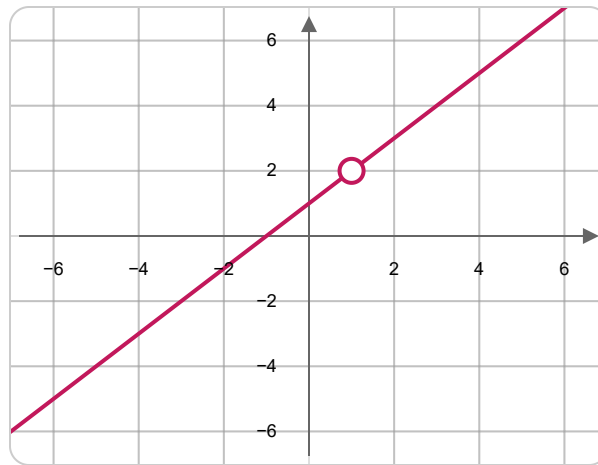
Hole: $(x-1)$ cancels $\rightarrow$ Hole at $x = 1$. Reduced: $x + 1$. $f(1) = 1 + 1 = 2 \rightarrow$ **Hole at** $(1, 2)$

VA: No remaining denominator $\rightarrow$ **No VA**

HA / SA: Simplified to line $y = x + 1 \rightarrow$ **No HA. Line with hole.**

x-int: $x + 1 = 0 \rightarrow (-1, 0)$

Hole: $(1, 2)$ | **No VA** | **Simplified:** $y = x + 1$



Line $y = x + 1$ with hole at $(1, 2)$

Problem 9

$$f(x) = \frac{x^2 + 2x + 1}{x + 1}$$

Perfect Square

Hole

No VA

Factor: $\frac{(x + 1)^2}{(x + 1)}$

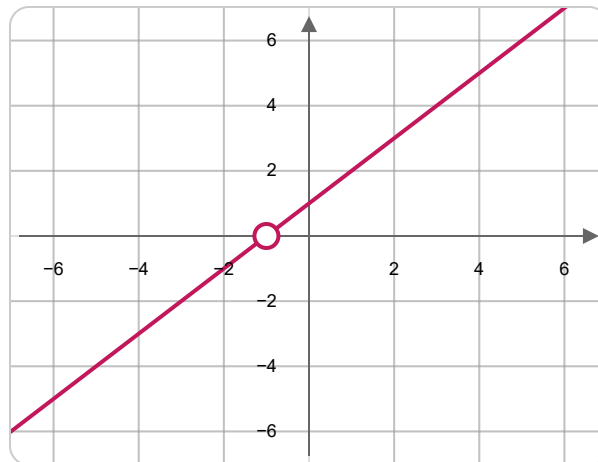
Hole: One $(x + 1)$ cancels $\rightarrow$ Hole at $x = -1$. Reduced: $x + 1$. $f(-1) = -1 + 1 = 0 \rightarrow$ **Hole at $(-1, 0)$**

VA: No remaining denominator $\rightarrow$ **No VA**

HA / SA: Simplified to line $y = x + 1 \rightarrow$ **No HA. Line with hole at $(-1, 0)$.**

Note: $(x + 1)^2 / (x + 1) = x + 1$ for $x \neq -1$. The hole is right on the x -axis since $y(-1) = 0$.

Hole: $(-1, 0)$ | **No VA** | **Simplified:** $y = x + 1$



Line $y = x + 1$ with hole at $(-1, 0)$ (on the x -axis)

Problem 10

$$f(x) = \frac{x^2 - 2x - 3}{x - 3}$$

Hole

No VA

Linear: $y=x+1$

Factor: $\frac{(x-3)(x+1)}{(x-3)}$

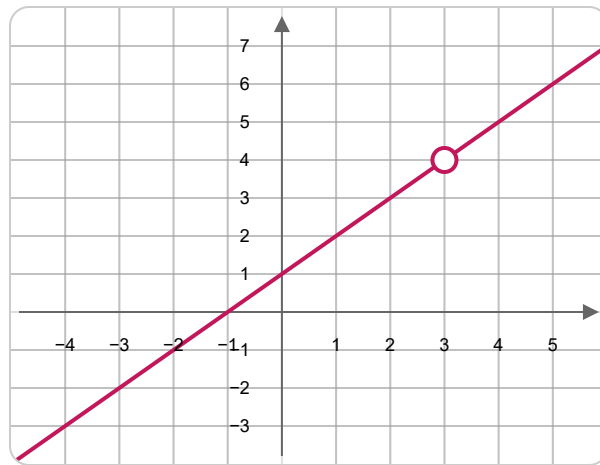
Hole: $(x-3)$ cancels $\rightarrow$ Hole at $x=3$. Reduced: $x+1$. $f(3) = 3+1=4 \rightarrow$ **Hole at $(3, 4)$**

VA: No remaining denominator $\rightarrow$ **No VA**

HA / SA: Simplified to line $y = x + 1 \rightarrow$ **No HA. Line with hole.**

x-int: $x+1=0 \rightarrow (-1, 0)$

Hole: $(3, 4)$ | **No VA** | **Simplified:** $y = x + 1$



Line $y = x + 1$ with hole at $(3, 4)$